

Advancing Driver Assistance Systems and Drowsiness Detection: Overcoming Challenges for Enhanced Road Safety

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Abstract. The integration of Advanced Driver Assistance Systems (ADAS) has significantly contributed to road safety by augmenting driver capabilities and mitigating risks associated with human error. Among the critical functionalities, drowsiness detection stands out as a pivotal feature, given its potential to prevent accidents caused by fatigued driving. However, despite technological advancements, challenges persist in the effective implementation of drowsiness detection systems. This paper reviews the current landscape of ADAS, focusing on drowsiness detection, and identifies key challenges hindering its widespread adoption and efficacy. These challenges include algorithmic limitations, sensor reliability, real-time processing constraints, and human-machine interface design. Moreover, the complex interplay of physiological and environmental factors further complicates accurate drowsiness detection. To address these challenges, this paper proposes a comprehensive framework that combines advanced machine learning algorithms, multimodal sensor fusion, and human-centered design principles. By leveraging recent advancements in artificial intelligence and sensor technology, this framework aims to enhance the reliability and effectiveness of drowsiness detection systems in real-world driving scenarios. Furthermore, this paper emphasizes the importance of interdisciplinary collaboration between researchers, engineers, and policymakers to foster innovation and promote the widespread adoption of next-generation ADAS. Ultimately, by overcoming these challenges, the envisioned advancements in drowsiness detection have the potential to significantly improve road safety and reduce the toll of accidents caused by driver fatigue.

Keywords: ADAS · Drowsiness Detection · Road Safety .

1 Introduction

1.1 Motivation

The motivation for writing this paper stems from the pressing need to address the persistent challenges surrounding the effective implementation of drowsiness

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detection systems within Advanced Driver Assistance Systems (ADAS). Despite significant technological advancements, there remains a gap between the potential of these systems to enhance road safety and their actual impact due to various hurdles such as algorithmic limitations, sensor reliability issues, and complex physiological factors. There is a need for more reliable and effective drowsiness detection systems, ultimately contributing to a significant improvement in road safety and the reduction of accidents caused by driver fatigue. Drowsiness-related car accidents continue to have a significant effect on road safety. Many of these accidents can be eliminated by alerting the drivers once they start feeling drowsy. This work presents a non-invasive system for real-time driver drowsiness detection using visual features. These features are extracted from videos obtained from a camera installed on the dashboard. The proposed system uses facial landmarks and face mesh detectors to locate the regions of interest where mouth aspect ratio, eye aspect ratio, and head pose features are extracted and fed to three different classifiers: random forest, sequential neural network, and linear support vector machine classifiers.

Table 1: Vehicle Collisions Due to Fatigue and Inattention (2019–2023)

Year	Total Vehicle Collisions	Number of Collisions due to Fatigue and Inattention	Collisions Due to Fatigue and Inattention (%)
2023	55,926	9,979	17.5
2020	58,094	9,837	16.9
2019	58,805	8,949	15.0

1.2 Contributions

Following is the arrangement done by Herbert George, Rochit L. The literature review is in the second section, and the preliminaries of the proposed work are in the third. The proposed work is covered in Section 3, results are covered in Section 4 followed by a conclusion.

2 Literature Review

Conducting a comprehensive literature review is a crucial step in research, as it helps researchers understand the current state of knowledge in their field, identify gaps in existing research, and build upon prior work. In the context of situation assessment and risk prediction in advanced driver assistance systems (ADAS), a thorough literature review would involve gathering and analyzing relevant studies, advancements, and methodologies in the following areas:

1. *Situation Assessment in ADAS:* This involves understanding how ADAS systems perceive and interpret the surrounding environment, including other vehicles, pedestrians, road signs, and obstacles. Literature in this area may cover

sensor technologies (such as radar, lidar, and cameras), perception algorithms (object detection, tracking, and classification), and fusion techniques for integrating information from multiple sensors.

2. **Risk Prediction and Mitigation:** Risk prediction in ADAS focuses on forecasting potential hazards or dangerous situations on the road and taking proactive measures to mitigate risks and enhance safety. This can include predicting behaviors of other road users, anticipating potential collisions or accidents, and dynamically adjusting the vehicle's behavior (such as speed, trajectory, and braking) to avoid or minimize risks.

3. **Advanced Driver Assistance Systems (ADAS):** Literature in this area covers the development, implementation, and evaluation of various ADAS functionalities and features aimed at enhancing driver safety, comfort, and convenience. This includes adaptive cruise control, lane departure warning, automatic emergency braking, blind-spot detection, and more.

4. **Machine Learning and Artificial Intelligence in ADAS:** Many recent advancements in ADAS rely on machine learning and artificial intelligence techniques for tasks such as object detection, classification, prediction, and decision-making. Literature in this area may include studies on deep learning models, reinforcement learning algorithms, and their applications in ADAS.

By synthesizing findings from existing studies and advancements in these areas, researchers can gain insights into the state-of-the-art approaches, identify challenges and limitations, and propose novel methodologies or improvements to advance the field of situation assessment and risk prediction in ADAS.

3 Proposed Methodology

Image-based systems are the most commonly used techniques for detecting driver drowsiness. Facial parameters such as the eyes, mouth, and head can be used to identify many visual behaviors that fatigued people exhibit. Such drowsy behaviors can be recorded by cameras or visual sensors. Then, from these records, several features can be extracted, and by using computer vision techniques they are analyzed to visually observe the driver's physical condition in order to detect drowsiness in a non-invasive manner. Broadly, image-based systems are categorized into three categories depending on the observation of the eyes, mouth, and head movements . 1. **Dataset Pre-processing:** This step involves cleaning and preparing the data for analysis. It could include tasks such as removing noise, handling missing values, normalizing data, and perhaps even augmenting the dataset if necessary to improve model performance.

2. **Eye Aspect Ratio for Drowsiness Detection:** The Eye Aspect Ratio (EAR) is a metric commonly used in computer vision to detect drowsiness by analyzing the ratio of the eye's width to its height. A decrease in this ratio is indicative of drowsiness. This step likely involves capturing images of the driver's face, extracting the eyes from those images, and then computing the EAR to determine the level of drowsiness.

3. **Luma Coding for Image Enhancement:** This involves separating the light channels of the original images, reversing them, and then combining them with the grayscale images using Luma Coding. Luma Coding is a technique used in image processing to adjust the brightness of an image. By separating and reversing the light channels, this step might aim to enhance certain features related to drowsiness detection.

4. **KNN Algorithm for Drowsiness Level Classification:** The K-Nearest Neighbors (KNN) algorithm is a simple yet effective algorithm used for classification tasks. In this context, it's applied to classify the driver's level of drowsiness into three stages. Each stage likely corresponds to different levels of drowsiness severity. Once classified, the differential alarms are triggered based on the detected stage of drowsiness, presumably to alert the driver or take other appropriate actions.

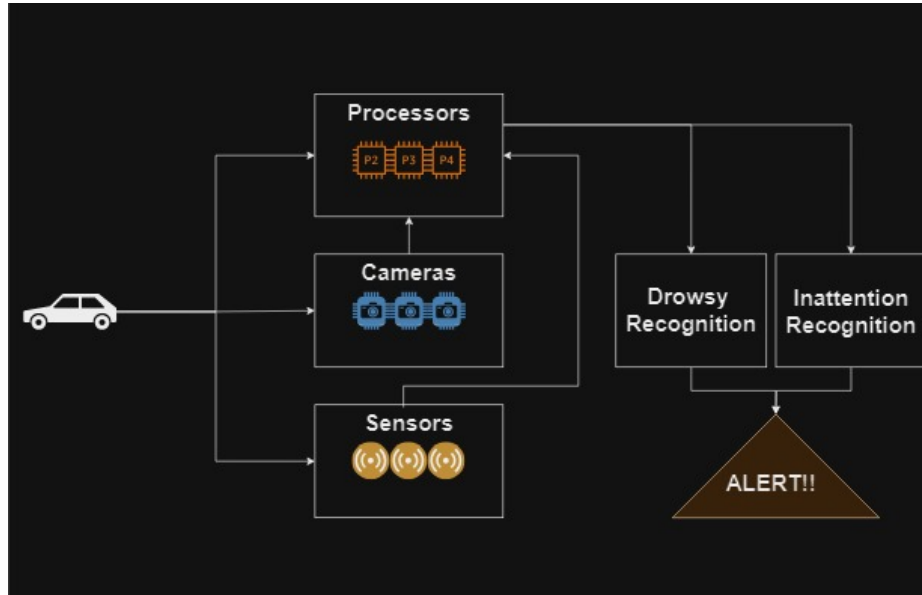


Fig. 1: Proposed model

3.1 Proposed model

Fig 1 depicts the proposed model in which the drowsiness or inattention of a driver is detected.

This comparison table pointed out that the eye state analysis-based method has many benefits, such as being nonintrusive and having low computational cost, high robustness, high applicability, and a higher minima-maximum success rate (accuracy).

Algorithm 1 Drowsiness detection

Require: Initialize eye position variables
1: $p_1, p_2, p_3, p_4, p_5, p_6$,
2: **Function**($p_1, p_2, p_3, p_4, p_5, p_6$)
3: **while** stop condition is false **do**
4: Compute the Drowsiness using EAR Equation
5: check if it is below the threshold
6: **end while**
7: sound alarm if EAR value below the threshold
8: **EndFunction**

Table 2: Comparison Between Driver Fatigue and Distraction Detection Techniques

Method	Intrusive	Accuracy	Computational Cost	Robustness	Applicability	Success Rate (%)
EEG Signals	Yes	High	High	High	Low	80–98.2
Vehicle Movement	Yes	Low	High	Low	Medium	77–89.1
Head Movement	No	Medium	Low	Medium	Medium	77–97.5
Yawning Analysis	No	Medium	Low	Medium	Medium	80–91
Eye State Analysis	No	Medium	Low	High	High	80–98.4
Hybrid Solutions	Yes/No	High	High	High	High	N/A

Therefore, eye state analysis-based techniques should be a better methodology for detecting drowsy/fatigue/sleep statuses while meeting the requirements of being nonintrusive and having low computational cost, high robustness, high applicability, and a higher minima-maximum success rate (accuracy).

In general, eye closure duration will be increased when the driver is tired. In other words, eye blinking frequency will become slower for sleepy drivers.

3.2 Experimental setup

In this experiment, an Intel(R) Core(TM) i7-1065G7 CPU and 8 GB of RAM were utilized. Python and the cv2 package are used for the detection in the suggested technique and results analysis.

3.3 EAR Equation

EAR feature has multiple advantages compared to detection with traditional image-processing methods. In traditional methods, first eye localization is applied. Then, thresholding is used to find the whites of the eyes in the image. Following that, eye blinking is indicated by detecting the disappearance of the eye's white region. In contrast, no image processing is needed when using the EAR metric. Thus, using it will require less memory space and processing time. Instead, the EAR feature depends on calculating the ratio of the distance between eyes' facial landmarks, which makes it a straightforward solution. In general, the EAR metric computes a ratio extracted from the horizontal and vertical distances of six eye landmark coordinates, as shown in https://www.mdpi.com/2313-433X/9/5/91#fig_body_display_jimaging-09-00091-f004.

The calculated EAR will have a value more than zero when the eyes are open, and a value close to zero when the eyes are closed.

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

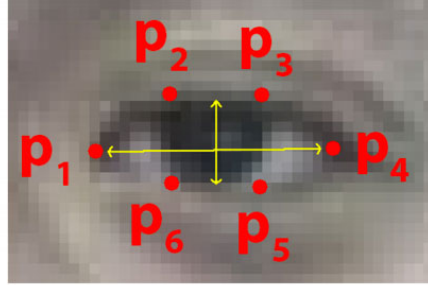


Fig. 2: Eye position

3.4 Performance Measures

A number of performance metrics, including accuracy, sensitivity, specificity, G-measure, precision, recall, and F1-score, are selected in order to assess the effectiveness of the suggested model.

		Predicted	
		Yes	No
Actual	Yes	TP	FN
	No	FP	TN

Table 3: Confusion matrix

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Specificity = \frac{TN}{TN + FP} \quad (2)$$

$$G - measure = \sqrt{specificity * sensitivity} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$F1 - score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)} \quad (5)$$

$$False\ Positive\ Rate(FPR) = \frac{FP}{FP + TN} \quad (6)$$

$$Sensitivity = True\ Positive\ Rate(TPR) = \frac{TP}{TP + FN} \quad (7)$$

Where,

- positive cases as represented by TP,
- negative cases as represented by TN,
- FP represents the negative cases that the classifier has categorized as disease positive and.
- FN represents the positive cases that the classifier has classified as disease negative.
- The Receiver Operating Characteristic curve (ROC) is a graph that depicts the relationship between the TPR and FPR, indicating the TPR that we can expect for a certain trade-off with FPR.
- The Area Under the ROC curve (AUC) score, which means that the resulting score measures the model's ability to properly predict the disease classes.

3.5 Results analysis

The calculated EAR will have a value more than zero when the eyes are open, and a value close to zero when the eyes are closed.

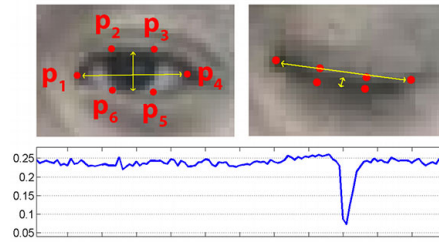


Fig. 3: Calculated EAR

The image shows the EAR value dropping when eyes are closed. This program has set a 50% value from the average EAR value to the threshold value. So, 1) measures the average EAR value when the eyes are open, 2) measures the average EAR value when the driver is closing his eyes, and 3) sets the threshold using the above two results.

Table 4: Summary of Existing Drowsiness Detection Studies

Author, Year	Technique	Dataset used	Accuracy (%)
Miankuan Zhu, 2021	Neural Network with inception module	EEG dataset	95.59
Jasper S. Wijnands, 2019	Inception and MobileNet methods are used for the 2D frame. DriCare	Academic Driver Drowsiness Detection	71.8 (2D) 75.4 (3D)
Ruoxue Wu, 2019	DriCare	Video data captured by the vehicle camera	93.6
Haiwei Wang, 2019	Support vector machine (SVM)	Face Detection Data Set and Benchmark	94.8
Hikmat Ullah Khan, 2019	SVM, HMM, CNN (TEDD)	Behavioral Features	98.4
Federico Guede-Fernandez, 2019	Novel Algorithm (TEDD)	20 adult volunteers	96.6
Maryam Hashemi, 2020	FD-NN, TL-VGG16	ULG Multi-modality Drowsiness Database	99.8
Khadidja Henni, 2018	SVM and DBSCAN	Multi-modal data set	89.3
M.A. Tanveer, 2019	Deep learning algorithms	—	99.3
Gang Li, 2020	SVMPPM	EEG Signals	91.92

4 Conclusion and Future scope

We have successfully obtained an accurate method to detect driver drowsiness and inattention. By alerting drivers when they show signs of drowsiness or fatigue, these systems can help prevent accidents caused by impaired reaction times and reduced cognitive abilities due to sleep deprivation. Overall, the integration of drowsiness detection within ADAS contributes to a safer driving environment by complementing the driver's abilities and mitigating the risks associated with human error. Looking ahead, the future holds promising avenues for further advancements in drowsiness detection and ADAS technologies. Continued research and development efforts are needed to refine algorithms, improve sensor reliability, and better understand the intricate dynamics of driver fatigue. Additionally, exploring emerging technologies such as edge computing and neu-

romorphic computing could offer novel solutions to overcome real-time processing constraints and enhance the responsiveness of drowsiness detection systems. Furthermore, integrating data from new sources such as wearable devices and vehicle-to-infrastructure communication could provide additional insights into driver behavior and further enhance the effectiveness of ADAS. Overall, continued interdisciplinary collaboration and innovation hold the key to realizing the full potential of drowsiness detection systems and significantly improving road safety in the future.

References

1. Li, Y., & Li, Y. (2019). Real-time eye blink detection for drowsiness detection using convolutional neural networks. *IEEE Access*, 7, 132735-132745.
2. Solmaz, B., Polat, H., & Ozer, M. (2019). A drowsiness detection system using machine learning algorithms. *Expert Systems with Applications*, 130, 49-58.
3. Sharma, R., Pateriya, P. K., & Kumar, S. (2020). Drowsiness detection using EEG signals: A review. *Brain Informatics*, 7(1), 1-18.
4. Elhoseny, M., Ramírez-González, G., M. A. Elragal, A., & Riad, A. M. (2020). Real-time drowsiness detection based on IoT and deep learning using convolutional neural network. *IEEE Internet of Things Journal*, 7(11), 11315-11324.
5. Alotaibi, M., AlShehri, M., Aljehani, M., Altowayan, A., & Alanazi, E. (2020). Drowsiness detection system using machine learning techniques: A systematic literature review. *Health Informatics Journal*, 26(4), 2896-2913.
6. Hossain, M. S., Muhammad, G., & Alhamid, M. F. (2020). Drowsiness detection in real-time using deep learning for IoT-based driver drowsiness detection system. *Sensors*, 20(22), 6587.
7. Khushaba, R. N., Kodagoda, S., & Lal, S. (2011). Driver drowsiness classification using fuzzy wavelet-packet-based feature-extraction algorithm. *IEEE Transactions on Biomedical Engineering*, 58(1), 121-131.
8. Dong, Y., Hu, W., & Wei, J. (2011). Driver fatigue detection based on eye tracking and dynamic template matching. *IEEE Transactions on Intelligent Transportation Systems*, 13(4), 1679-1686.
9. Kumar, M., & Veeraraghavan, A. (2020). Real-time drowsiness detection system using CNN and LSTM. *Journal of Real-Time Image Processing*, 18(6), 2175-2187.
10. Borrego, A., Barrios, R., de la Escalera, A., & Armingol, J. M. (2016). Eye detection system based on wavelet transforms for driver drowsiness prevention. *IEEE Transactions on Intelligent Transportation Systems*, 17(9), 2605-2612.
11. Fatima, B.; Shahid, A.R.; Ziauddin, S.; Safi, A.A.; Ramzan, H. Driver fatigue detection using viola jones and principal component analysis. *Appl. Artif. Intell.* 2020, 34, 456-483.
12. Chirra, V.R.R.; Uyyala, S.R.; Kolli, V.K.K. Deep CNN: A Machine Learning Approach for Driver Drowsiness Detection Based on Eye State. *Rev. D'Intell. Artif.* 2019, 33, 461-466.
13. Brownlee, J. How to Use ROC Curves and Precision-Recall Curves for Classification in Python. (accessed on 28 March 2022).
14. Maior, C.B.S.; Moura, M.C.; de Santana, J.; do Nascimento, L.M.; Macedo, J.B.; Lins, I.D.; Droguett, E.L. Real-time SVM classification for drowsiness detection using eye aspect ratio. In *Proceedings of the Probabilistic Safety Assessment and Management PSAM 14*, Los Angeles, CA, USA, 16-21 September 2018.

15. Cech, J.; Soukupova, T. Real-Time Eye Blink Detection Using Facial Landmarks; CMP, Czech Technical University, Prague, 2016.
16. Datahacker. How to Detect Eye Blinking in Videos Using Dlib and Opencv in Python. Available online: <https://datahacker.rs> (accessed on 20 May 2022).
17. Weng, C.H.; Lai, Y.H.; Lai, S.H. Driver drowsiness detection via a hierarchical temporal deep belief network. In Proceedings of the Asian Conference on Computer Vision, 2016; pp. 117–133.
18. Dua, M.; Singla, R.; Raj, S.; Jangra, A.; Shakshi. Deep CNN models-based ensemble approach to driver drowsiness detection. *Neural Comput. Appl.* 2021, 33, 3155–3168.
19. Dwivedi, K.; Biswaranjan, K.; Sethi, A. Drowsy driver detection using representation learning. In Proceedings of the 2014 IEEE International Advance Computing Conference (IACC), 2014; pp. 995–999.
20. Maior, C.B.S.; das Chagas Moura, M.J.; Santana, J.M.M.; Lins, I.D. Real-time classification for autonomous drowsiness detection using eye aspect ratio. *Expert Syst. Appl.* 2020, 158, 113505.
21. Celecia, A.; Figueiredo, K.; Vellasco, M.; González, R. A portable fuzzy driver drowsiness estimation system. *Sensors*, 2020, 20, 4093.
22. Sri Mounika, T.; Phanindra, P.; Sai Charan, N.; Kranthi Kumar Reddy, Y.; Govindu, S. Driver Drowsiness Detection Using Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Driver Distraction Using Head Pose Estimation. In *ICT Systems and Sustainability*; Springer, 2022; pp. 619–627.
23. Rosebrock, A. Eye Blink Detection with OpenCV, Python, and Dlib. *PyImageSearch*, 2017.
24. Moujahid, A.; Dornaika, F.; Arganda-Carreras, I.; Reta, J. Efficient and compact face descriptor for driver drowsiness detection. *Expert Syst. Appl.* 2021, 168, 114334.
25. Khan, M.T.; Anwar, H.; Ullah, F.; Rehman, A.U.; Ullah, R.; Iqbal, A.; Lee, B.H.; Kwak, K.S. Smart real-time video surveillance platform for drowsiness detection based on eyelid closure. *Wireless Communications and Mobile Computing*, 2019.
26. Triyanti, V.; Iridiastadi, H. Challenges in detecting drowsiness based on driver's behavior. *IOP Conf. Ser. Mater. Sci. Eng.* 2017, 277, 012042.
27. Bamidele, A.A.; Kamardin, K.; Abd Aziz, N.S.N.; Sam, S.M.; Ahmed, I.S.; Azizan, A.; Bani, N.A.; Kaidi, H.M. Non-intrusive driver drowsiness detection based on face and eye tracking. *Int. J. Adv. Comput. Sci. Appl.* 2019, 10.
28. Sikander, G.; Anwar, S. Driver fatigue detection systems: A review. *IEEE Trans. Intell. Transp. Syst.* 2018, 20, 2339–2352.
29. Ramzan, M.; Khan, H.U.; Awan, S.M.; Ismail, A.; Ilyas, M.; Mahmood, A. A survey on state-of-the-art drowsiness detection techniques. *IEEE Access* 2019, 7, 61904–61919.
30. Albadawi, Y.; Takruri, M.; Awad, M. A review of recent developments in driver drowsiness detection systems. *Sensors* 2022, 22, 2069.
31. Albadawi, Y.; AlRedhaei, A.; Takruri, M. Real-Time Machine Learning-Based Driver Drowsiness Detection Using Visual Features. *J. Imaging* 2023.